

Weather Near-Resolution Lag-Logger

A read-only measurement instrument for Polymarket weather markets — Contest Edge (contest-edge).
Generated 22 Jul 2026.

What it is

It is **not a trading bot and not a pick-seller**. It is an instrument that answers one question with evidence: when a Polymarket weather market becomes *mechanically decided* by the weather that has already been observed, how long does the market price take to catch up — and how many dollars are actually fillable in that gap? It places no orders, holds no wallet, and uses no API keys. It only reads and logs.

The core idea: a fact, not a forecast

Polymarket runs markets like “Highest temperature in Beijing on July 22?” with buckets (e.g. 73–74°C, 75°C or above). Each market settles on the highest temperature recorded that day at one specific airport weather station (via Wunderground).

A daily maximum can only rise. So the moment the station’s observed high for the day clears a bucket’s boundary, that bucket’s outcome is **settled in advance** — it is a fact, no longer a prediction:

- **PROVEN** — the station already reached the floor of an “X or above” bucket → that bucket resolves YES, forever.
- **DEAD** — the station already exceeded a bucket’s ceiling → that bucket resolves NO, forever.

The tool flags **only** these already-happened facts. “The peak has probably passed” is still a forecast and is deliberately never counted. A safety margin (1°F / 0.5°C) absorbs rounding differences between the raw station report and Wunderground’s published figure.

How it works — the pipeline

1. Rules parser	Reads each market’s own rules text and extracts the exact settlement station (ICAO code), the unit, and the source. Self-configuring — no hardcoded city list. Any market whose station can’t be identified is EXCLUDED and never watched.
2. Observation feed	Pulls the station’s real METARs from aviationweather.gov (free, global, no key) and tracks the running observed daily max since local midnight.
3. Local-day math	Uses a DST-correct UTC offset (from Open-Meteo) so “today” is the station’s local calendar day, not the server’s.
4. Lock detector	Compares the observed max to each bucket boundary (plus the safety margin) and marks PROVEN / DEAD / still-live.
5. Order-book snapshot	For a locked bucket, reads the public CLOB order book and computes the fillable EDGE-DOLLARS: how much could actually be bought at prices better than the now-certain value, walking the book.

6. Append-only log	Writes LOCK (first detection), SNAPSHOT (each later poll), and CONCEDE (price finally crosses 0.95/0.05) rows. Never mutates. The lag between LOCK and CONCEDE is the entire measurement.
7. Digest & page	Per-city verdict lines (lock count, median lag, median fillable \$) and an honest global ceiling, on a /triggers web page and API.

Why this market and not crypto

Short-horizon crypto up/down is a binary bet on a near-random coin flip the market already prices at ~50/50 — there is no structural edge to harvest, which is why our own ledger found model $\approx$ market. Weather near-resolution is the opposite: the settlement instrument has already printed the answer, and the only question is whether the retail crowd has updated. That is a mechanical lag, not a prediction — the same shape as the “both teams in the bonus” live-betting trigger.

What it deliberately does NOT do

- No orders, no wallet, no keys — strictly read-only.
- No picks and no forecasting — only already-observed facts.
- No claim of profit — it measures whether an edge exists.

Honest status (as of first run)

The instrument is built and validated: mechanics were cross-checked against raw London City Airport data (it really hit 25°C). A first snapshot flagged ~98 locks across 32 cities with a large notional “fillable” figure — but that number is a **hypothesis, not an edge**, for three reasons:

- No lag is measured yet — a single snapshot can’t show whether prices actually concede, and the lag IS the edge.
- The safe, deep-in-the-money buckets are already priced correctly (\$0 edge); the dollars concentrate on knife-edge buckets that carry the most rounding/revision risk.
- Observed-max correctness has been spot-checked on only 1 of 32 stations.

Next step: run it for several days to measure real concede-lag, and hand-audit a sample of locks against Wunderground before believing any dollar figure.

How to run it

```
python scan.py weather-rules # write the rules-review file
python scan.py weather-trigger once # a single read-only pass
python scan.py weather-trigger 1440 # watch loop, 24h, adaptive poll
```

Web: /triggers and /api/triggers on the ledger site. Data lands in the append-only trigger_events table.